UNIVERSITY OF CAPE TOWN

DEPARTMENT OF ELECTRICAL ENGINEERING

VACATION WORK REPORT

Microgrid Monitoring, Industrial Communication and Protection-System Integration

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Submitted in fulfilment of the practical-training requirements for EEE3000X.

Note: Employer confirmation showing the start and finish dates must be attached before submission.

1. Introduction and objectives

I completed six weeks of practical training in the UCT AEPEC laboratory, where I worked on an existing low-voltage microgrid used for teaching and research. My work concentrated on the parts of the system that measure electrical quantities, communicate those values to the operator interface, execute control commands and provide protection during abnormal operating conditions. The practical work therefore combined power-system measurement, PLC/HMI integration, industrial communication, protection relays and software-based data acquisition.

My main objective was to understand the complete path from a physical current or voltage in the microgrid to the value displayed or recorded by a computer. I also had to diagnose faults in that path and develop a proof-of-concept method for remotely extracting and logging measurements. The work was carried out under supervision, with guidance from Dr Maysam Soltanian and laboratory staff.

The specific objectives were to:

- identify the electrical and communication architecture of the microgrid;
- understand the use of current transformers (CTs), voltage inputs and DIRIS A-40 power meters;
- extract and analyse the Delta DVP28SV PLC and DOP-B07E415 HMI programs;
- map HMI objects and registers to PLC control points and meter measurements;
- troubleshoot Modbus RTU communication faults on the two-wire RS485 network;
- investigate the SEL-751A, SEL-710 and SEL-487E protection system and IEC 61850 GOOSE communication; and
- develop and evaluate a Python-based Modbus TCP monitoring and logging application.

2. Equipment and principles involved

The monitoring system used eleven Socomec DIRIS A-40 power meters connected to current transformers and voltage-measurement points. The meters converted the CT secondary currents and voltage inputs into engineering quantities such as phase current, phase and line voltage, active power, reactive power, apparent power, energy and power factor. Correct CT ratios, phase association and polarity were essential because an incorrect connection could produce inaccurate magnitude or phase information.

The meters communicated with a Delta DOP-B07E415 HMI over a two-wire RS485 network using Modbus RTU. Each meter had a unique station address, allowing the HMI to poll several devices on the same daisy-chained bus. A Delta DVP28SV PLC executed ladder logic for contactor control and exchanged commands and status information with the HMI. The protection layer used SEL-751A, SEL-710 and SEL-487E relays. These relays relied on direct electrical measurements for protection and used IEC 61850, including GOOSE messages, for high-speed peer-to-peer signalling.

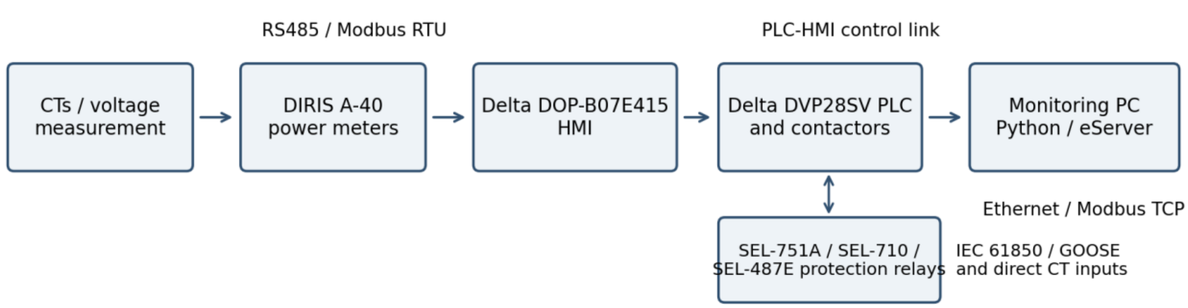

Figure 1. Simplified architecture investigated during the vacation work.

3. Work performed and implementation

3.1 Investigation of the electrical and control architecture

I first traced the physical system from the electrical feeders and busbars to the CTs, voltage inputs, power meters, HMI, PLC, contactors and protection relays. I studied the DIRIS A-40 documentation to understand how the configured network type and CT ratio affected the displayed quantities. This was necessary because the same communication value is only meaningful when it is associated with the correct phase, CT and scaling factor.

I then extracted the PLC program using WPLSoft and a USB-to-RS485 programming connection. I analysed the ladder logic to identify how input signals, HMI commands and interlocks controlled the contactors. I also extracted the HMI project through Ethernet using DOPSoft 2.00.07. This required resolving software-version, driver, IP-address and communication-setting problems before the project could be transferred successfully.

After obtaining both programs, I created an Excel mapping that linked HMI pushbuttons and numerical displays to PLC memory locations and meter registers. The mapping allowed me to connect a software object, such as a line-control button or current display, to the corresponding physical contactor or meter. I also added and tested selected numerical displays using the DIRIS register map. This work showed that the HMI was both an operator interface and the main Modbus master responsible for polling the meters.

3.2 Modbus RTU and HMI communication troubleshooting

A persistent HMI communication alarm led me to investigate the RS485 network at both the physical and application layers. I checked the two-wire A/B connections, device station numbers, baud-rate and serial settings, the daisy-chain topology, termination resistance, shield treatment and the routing of the communication cable near high-current conductors. I also investigated the meaning of HMI communication errors encountered during reading and tested whether the fault followed a specific meter, screen object or register.

The investigation showed that not every apparent network fault was caused by the cable. One problematic numerical display and its associated register contributed to the persistent error. The power-factor value was particularly unreliable, so I implemented an alternative calculation using active power divided by apparent power. A background macro was initially too slow, sometimes taking minutes to update. I replaced it with a clock macro running at a shorter interval, reducing the visible response to approximately ten seconds. This improved the display but also exposed the limited processing speed of the legacy HMI.

3.3 Remote monitoring and data logging

I next developed a proof-of-concept application to extract measurements from the HMI over Ethernet. Direct polling initially produced no response. I inspected the Ethernet communication settings in DOPSoft and created the required communication link. This linked the logical device/register path used in the project to the HMI Ethernet interface, after which the HMI responded to the Python Modbus TCP client.

Because the external computer could not directly read the remote DIRIS registers through the existing configuration, I used selected HMI internal registers as an intermediate data layer. A clock macro copied meter values into these registers, and the Python application read them through Modbus TCP, converted the raw values using the correct register type and scale, displayed them, and wrote time-stamped records to CSV/Excel-compatible files. The code was developed in Visual Studio Code and structured so that it could be shared through the group GitHub repository.

The proof of concept worked, but the practical update rate was approximately ten seconds. I compared this with Delta eServer, which obtained values more efficiently through Delta proprietary functionality. I concluded that the Python method was useful for understanding and custom development, while eServer was the more practical acquisition layer when faster logging was required from the existing HMI.

3.4 Protection-system investigation

I investigated the SEL protection system using SEL configuration tools and the relay front-panel indications. The SEL-751A and SEL-710 could be reached, while access to the SEL-487E IEC 61850 configuration was restricted by an authorisation/password problem. The relays also displayed GOOSE-related warnings, including messages indicating that expected subscriptions were not being received. This limited the ability to verify and modify the complete protection coordination.

I evaluated whether current values already available at the HMI could be sent to the protection relays to reduce wiring. I rejected this arrangement because the path CT - meter - HMI - protocol conversion - relay would introduce delay, dependence on several devices and a lower update rate. Protection relays require continuous and deterministic measurements, so their direct CT inputs should be retained. I also investigated the Moxa MGate 5119 as a possible protocol gateway, but it did not remove the fundamental measurement-latency problem and was not justified for the intended task.

During energisation tests, I helped investigate unexpected protection behaviour. The SEL-751A indicated a phase overcurrent trip even when only voltage was expected, while the SEL-487E indicated A- and C-phase differential elements although the displayed current magnitudes were zero. A phase-angle mismatch was also observed. In a separate fault, connecting the neutral associated with the C15/Load 1 measurement circuit caused the upstream earth-leakage protection to trip. I isolated the condition to that neutral connection and documented it as an unresolved neutral/earth-reference or secondary-wiring issue requiring controlled continuity, insulation, polarity and earthing checks before further energisation. I did not treat the neutral as a balancing conductor without first considering the complete earthing and protection arrangement.

4. Results obtained and engineering conclusions

Work area	Result obtained	Conclusion
PLC/HMI analysis	Programs were extracted and the control/register relationships were documented.	The mapping created a maintainable connection between software addresses and the physical microgrid.
RS485/Modbus	Physical-layer checks were completed and a problematic display/register was identified.	Troubleshooting must include wiring, protocol settings and the HMI application, not only the bus cable.
Python monitoring	The HMI responded over Modbus TCP and selected values were logged to CSV/Excel-compatible files.	The method is functional but constrained by the HMI macro and scan/update rate.
SEL protection	Relay access, GOOSE status and abnormal trips were investigated; some configuration access remained restricted.	Protection measurements should remain direct and deterministic; configuration backups and authorised access are essential.

The main technical outcome was a working end-to-end understanding of how measurements travelled from the microgrid to the HMI and then to an external computer. I produced documentation that can be used by the next student or technician, demonstrated independent Modbus TCP logging, and identified the practical limits of the installed HMI. Not every protection fault was resolved within the six-week period, but the work narrowed the remaining faults and prevented unsafe assumptions from being accepted as explanations.

5. Use of engineering course material

The work applied material from several parts of the Mechatronics Engineering curriculum. Circuit and electrical-machines concepts were used when interpreting CTs, three-phase quantities, power factor, phase sequence and earth-leakage behaviour. Digital systems and embedded/control concepts were used when analysing PLC ladder logic, scan-based control and HMI macros. Communication material was applied to RS485 differential signalling, serial parameters, Modbus addressing, Ethernet and TCP/IP. Protection and power-system principles were used to distinguish monitoring data from protection-grade measurements and to interpret relay elements and IEC 61850 GOOSE status. Python programming and data-handling skills were used to implement register polling, scaling, error handling and CSV logging.

6. Recommendations

- Retain direct CT and voltage inputs to protection relays; do not route protection measurements through the HMI.
- Back up the SEL settings, IEC 61850 configuration and access credentials, and maintain an authorised configuration record.
- Use Delta eServer as the acquisition layer for the current hardware, with a separate custom dashboard reading the logged data.

- For a long-term upgrade, replace the DOP-B07E415 with a modern HMI or dedicated industrial gateway that supports faster server and remote-access functions.
- Keep a controlled register map and network drawing showing station addresses, scaling, link assignments, termination points and shield/earth connections.
- Complete the C15 neutral/earth-leakage investigation under competent supervision before further live operation.

7. Conclusion

The vacation work gave me practical experience of a multidisciplinary industrial system rather than an isolated component. I moved from tracing CT and voltage measurements, to analysing PLC/HMI programs, mapping Modbus registers, diagnosing RS485 faults, accessing the HMI through Ethernet and developing a Python logging application. I also investigated SEL protection relays and learned why the speed, reliability and independence required by protection systems are different from the requirements of monitoring systems.

The project strengthened my ability to troubleshoot systematically. A communication alarm could originate from wiring, addressing, register type, a display object or a legacy-hardware limitation; similarly, an unexpected trip could involve measurement polarity, phase reference, earthing, wiring or IEC 61850 coordination. The final result was a documented microgrid architecture, a functional proof-of-concept monitoring system and a set of technically justified recommendations for safer and more maintainable operation.

References

1. Socomec, DIRIS A-40 Power Monitoring Device documentation and Modbus register tables.
2. Delta Electronics, DOP-B Series HMI and DOPSoft documentation.
3. Delta Electronics, DVP Series PLC and WPLSoft documentation.
4. Modbus Organization, Modbus Application Protocol Specification.
5. Schweitzer Engineering Laboratories, SEL-751A, SEL-710 and SEL-487E documentation.
6. IEC 61850 communication and GOOSE configuration documentation used in the laboratory.

Appendix A: Employer confirmation

Attach the signed letter, certificate or timesheet from the supervisor/organisation confirming the work completed and the exact start and finish dates. This appendix is required before the report is submitted.

Student signature: ________L.T_______________________ Date: ___03/08/2026______________

M. Soltanian 17/08/2026